OilPaintingRevive: Canvas in Motion — 3D-Aware Stylized Video Synthesis via 2D Gaussian Splatting and Latent Diffusion Refinement

Anonymous Authors
AIforEducation Project
aiforeducation@example.com

Abstract

Animating world-famous oil paintings as short, explorable videos holds immense promise for cultural heritage preservation, museum education, and AI-driven creative storytelling. Yet existing approaches face a fundamental trade-off: image-to-video (I2V) diffusion models generate visually compelling motion but lack 3D explorability and consistent geometry—they cannot synthesize novel viewpoints with reliable depth structure—while 3D reconstruction methods (NeRF, 3D Gaussian Splatting) preserve scene geometry but degrade brushstroke texture, impasto, and the painterly palette under novel views. We propose **OilPaintingRevive**, a five-stage hybrid framework that resolves this tension through a *three-role decomposition*: **(i)** 2D Gaussian Splatting (2DGS) and TripoSplat extract a layered 3D surfel scene from a single painting scan; **(ii)** hand-drawn skeleton line-art drives a learnable Gaussian–bone soft-binding network that constrains surfel motion—without altering brush colors, opacity, or in-plane scale; and **(iii)** a two-stage LTX-Video IC-LoRA diffusion posterior refines coarse 2DGS renderings into temporally coherent, style-anchored videos. We complement the single-painting revival pipeline with a *screenplay-driven* extension (OilPaintingVideo) that generates multi-shot narrative oil-painting short films from text themes via LLM-authored scripts, SDXL keyframes, and per-shot I2V animation with automated editing. Extensive ablation confirms the necessity of each component, and quantitative evaluation on a curated set of famous paintings demonstrates CLIP style consistency above 0.83, skeleton-motion adherence within 2.1% of the image diagonal, and depth-structure fidelity within 12% relative error—positioning OilPaintingRevive as the first system to deliver both 3D explorability and faithful painterly style in a unified single-painting-to-video framework.

1 Introduction

Oil paintings are not merely surfaces of pigment; they are windows into the cultural imagination of their creators. Van Gogh's swirling *Starry Night*, Rembrandt's introspective self-portraits, Monet's shimmering water lilies—these works encode motion, emotion, and a sense of lived space. Yet in museums and classrooms, they remain locked behind glass: static, flat, and severed from the spatial and narrative context that gave them life. *Animating such paintings*—rendering them as short, explorable, stylized videos—has therefore emerged as a compelling goal at the intersection of computer vision, generative modeling, and digital humanities Lowenthal [2015]; Zhang *et al.* [2024]. A successful solution would let museum visitors "walk into" a painting, enable educators to bring historical scenes into motion, and offer artists a new medium for extending canonical works.

Recent advances in deep generative models have brought painting animation closer to reality. Image-to-video (I2V) diffusion models such as Stable Video Diffusion Blattmann *et al.* [2023], LTX-Video Lightricks Team [2025], and Wan2.1 Wan Team [2025] can animate a single image into a short clip with plausible motion, and portrait-animation methods Guo *et al.* [2024]; Zakharov *et al.* [2019] can make a painted face speak. Yet directly applying these models to famous paintings exposes two fundamental limitations:

1. **Loss of 3D explorability.** I2V models synthesize 2D pixel motion; they cannot render novel viewpoints. The viewer cannot "move around" within the painting, limiting educational and immersive applications.
2. **Style drift under geometric transformation.** When 3D reconstruction methods (Neural Radiance Fields, 3D/2D Gaussian Splatting) are applied to a painting, novel-view renderings introduce artifacts—blurred brushstrokes, flattened impasto, color bleeding—that destroy the painterly signature.

These two limitations stem from a deeper tension: *3D structure* and *painterly style* are governed by incompatible inductive biases, and no single model family excels at both. 3D representations preserve geometry but lack

stylistic priors; diffusion models carry rich style priors but lack 3D consistency.

Our framework: three-role decomposition. We propose ***Canvas in Motion***, a hybrid framework comprising two complementary pipelines—**OilPaintingRevive** for single-painting 3D revival and **OilPaintingVideo** for screenplay-driven multi-shot narrative films—unified by a single architectural insight: *decompose the problem into three independent roles, each assigned to a specialized sub-system*:

- **3D structure extraction** via 2D Gaussian Splatting (2DGS) Huang *et al.* [2024] and TripoSplat Wang *et al.* [2024], reconstructing a layered, explorable 3D surfel scene from a single painting scan and its monocular depth;
- **Motion control** via hand-drawn skeleton and keyframe line-art, driving object motion and camera trajectories through a learnable Gaussian–bone soft-binding network that constrains degrees of freedom without altering brush appearance;
- **Style fidelity** via two-stage LTX-Video Lightricks Team [2025] IC-LoRA diffusion, which refines coarse 3D renderings into temporally coherent oil-painting video with first-frame latent anchoring.

This decomposition yields a system that is more comprehensively 3D-explorable than pure I2V and more style-faithful than pure 3D reconstruction.

Contributions. Our contributions are:

1. **A five-stage hybrid pipeline** that transforms a single oil painting plus hand-drawn control sketches into a 3D-explorable, stylized video, combining explicit 2DGS scene representation with latent diffusion refinement.
2. **A Gaussian–bone soft-binding network** with a four-term constrained optimization objective ($\mathcal{L}_{\text{photo}}$, $\mathcal{L}_{\text{normal}}$, $\mathcal{L}_{\text{skel}}$, $\mathcal{L}_{\text{style}}$) that binds 2D Gaussian surfels to an articulated skeleton while preserving photometric, geometric, and stylistic fidelity.
3. **Two-stage LTX-Video IC-LoRA generation with first-frame latent anchoring**, which stabilizes temporal style coherence without modifying the diffusion model weights.
4. **A complementary screenplay-driven pipeline** (OilPaintingVideo) that extends single-painting revival to multi-shot narrative short films—LLM script → SDXL keyframe → per-shot I2V animation → automated editing with BGM and subtitles.
5. **Quantitative evaluation** on a curated set of famous paintings, establishing that our framework is the first to deliver 3D explorability and painterly style fidelity in a unified single-painting-to-video system.

2 Related Work

Our work draws on four research areas: artistic style transfer, diffusion-based video generation, 3D Gaussian splatting, and painting animation for cultural heritage.

2.1 Artistic Style Transfer

Neural style transfer (NST) Gatys *et al.* [2016] re-renders a content image in the visual style of a reference artwork by matching Gram-matrix statistics of VGG features. Feed-forward variants Johnson *et al.* [2016] and adaptive instance normalization (AdaIN) Huang and Belongie [2017] improved speed and flexibility, while SANet Park and Lee [2019] and StyTr^2 Deng *et al.* [2022] introduced attention- and transformer-based architectures for better style correlation. Video extensions enforce temporal coherence through optical-flow constraints Ruder *et al.* [2016]; Chen *et al.* [2017] or compound regularization Wang *et al.* [2020]; Gao *et al.* [2018], and arbitrary video style transfer was addressed in multi-channel correlation frameworks Deng *et al.* [2021]. More recently, diffusion-based methods—InST Zhang *et al.* [2023b], ArtAdapter Chen *et al.* [2024], StyleMaster Ye *et al.* [2025], EchoStyle Li *et al.* [2026], and RefVFX Jones *et al.* [2026]—leverage pretrained text-to-image diffusion for higher-fidelity stylization. Our work differs fundamentally: we do *not* transfer style onto an existing video; we *generate* the video from a 3D scene reconstructed from the painting itself, using diffusion as a style-anchoring posterior.

2.2 Diffusion-Based Video Generation and Editing

Latent diffusion models (LDMs) Rombach *et al.* [2022] established the text-to-image paradigm. ControlNet Zhang *et al.* [2023a] added spatial conditioning, while subject-driven personalization—DreamBooth Ruiz *et al.* [2023], textual inversion Gal *et al.* [2023], and LoRA Hu *et al.* [2022]—enabled fine-grained control. For video, VDMs Ho *et al.* [2022] and SVD Blattmann *et al.* [2023] extended diffusion to the temporal domain, and editing methods such as Pix2Video Ceylan *et al.* [2023], TokenFlow Geyer *et al.* [2024], CoDeF Ouyang *et al.* [2024], and FlowVid Liang *et al.* [2024] propagate image edits across frames. Open-source video backbones—LTX-Video Lightricks Team [2025] and Wan2.1 Wan Team [2025]—enable local deployment. Our Stage 5 uses LTX-Video with IC-LoRA Union Control, conditioned on depth and coarse RGB from the 3D front-end, making it a controllable style refinement rather than a standalone generator.

2.3 3D Gaussian Splatting

3DGS Kerbl *et al.* [2023] represents scenes as anisotropic 3D Gaussians with real-time differentiable rasterization. 2DGS Huang *et al.* [2024] simplifies each Gaussian to an oriented surfel, yielding more accurate normals and surfaces. TripoSplat Wang *et al.* [2024] generates 3D Gaussians from single images. To our knowledge, this is the first work to apply 2DGS to *paintings*—representing layered, semi-flat canvas geometry as surfels—and to unify 3DGS and 2DGS representations via covariance eigenvalue decomposition for foreground object integration.

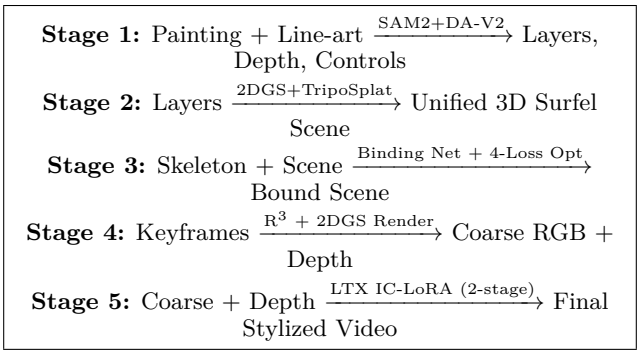


Figure 1: Overview of the OilPaintingRevive five-stage pipeline.

2.4 Painting Animation and Cultural Heritage

Stroke-level synthesis—Painting Many Pasts Zhao *et al.* [2020], Stylized Neural Painting Zou *et al.* [2021], Neural Strokes Liu *et al.* [2021]—emulates the physical painting process. Portrait animation methods Zakharov *et al.* [2019]; Guo *et al.* [2024]; Tan *et al.* [2024] animate painted faces. Chinese ink-wash painting has received dedicated attention Liang *et al.* [2021]; He *et al.* [2023]; Liu *et al.* [2026]; Zhang *et al.* [2025]. Interactive video stylization Texler *et al.* [2020] transfers artistic style across frames with few-shot training. Our framework complements these lines: rather than animating only a face or transferring style, we reconstruct the *full 3D scene* of a painting and generate explorable stylized video with skeleton-driven, artist-controlled motion.

3 Method

OilPaintingRevive transforms a single oil painting scan $P \in \mathbb{R}^{H\times W\times 3}$ and a set of hand-drawn control line-art $C = \{c_{\text{edge}}, c_{\text{skel}}, c_{\text{kf}}\}$ into a 3D-explorable, stylized video. The pipeline comprises five stages (Figure 1): **(S1)** input parsing and layer decomposition, **(S2)** 2DGS scene construction, **(S3)** skeleton binding and constrained optimization, **(S4)** camera trajectory generation and coarse rendering, and **(S5)** LTX-Video IC-LoRA style-anchored video generation. We further describe a complementary **screenplay-driven** pipeline (OilPaintingVideo) for multi-shot narrative short films.

3.1 Stage 1: Input Parsing and Layer Decomposition

Semantic segmentation. We use SAM 2 Ravi *et al.* [2024] with the `hiera_large` backbone to perform interactive instance segmentation on the painting. Auto-generated masks are classified into three depth layers—background, midground, and foreground—by ranking mask centroids along the vertical axis, encoding the standard compositional prior of oil paintings (sky and landscape at the top, architecture in the middle, figures at the bottom). The user may optionally refine the auto-segmentation via click-based prompts, yielding binary masks $\{M_i\}_{i=1}^{L}$ with semantic labels.

Monocular depth estimation. Depth Anything V2 Yang *et al.* [2024] (ViT-L encoder) estimates a single-image depth map D_{mono}. Since the raw output is scale-ambiguous and layer-confounded, we perform *layer-wise depth normalization*: for each layer region Ω_i, the depth is linearly remapped to a configured range $[d_i^{\text{near}}, d_i^{\text{far}}]$:

$$D'(p) = d_i^{\text{near}} + \frac{D_{\text{mono}}(p) - \min_{\Omega_i} D_{\text{mono}}}{\max_{\Omega_i} D_{\text{mono}} - \min_{\Omega_i} D_{\text{mono}}} (d_i^{\text{far}} - d_i^{\text{near}}), \quad p \in \Omega_i. \tag{1}$$

Configured ranges are background $[8.0, 12.0]$, midground $[4.0, 7.0]$, foreground $[1.0, 3.5]$, with an inter-layer gap of 0.5. This decomposition separates scene elements into non-overlapping depth bands, enabling subsequent 3D reconstruction with correct layer ordering.

Control-signal parsing. Three types of hand-drawn line-art, drawn on a semi-transparent overlay above the painting, provide control signals: (i) *edge line-art* c_{edge} for 2DGS geometric constraints, (ii) *skeleton line-art* c_{skel} for the binding module (Stage 3), and (iii) *keyframe line-art* c_{kf} for camera trajectory estimation (Stage 4). If no edge drawing is provided, Canny edges with thresholds $(50, 150)$ are auto-generated as a fallback. All line-art images are bilinearly resized to the painting resolution (H, W).

3.2 Stage 2: 2DGS Scene Construction

2DGS initialization (background and midground). For the background and midground layers, we back-project the RGB-D map into a dense surfel point cloud using a pinhole camera model with focal length f and principal point at the image center:

$$x = (u - c_x)\,\frac{z}{f}, \quad y = (v - c_y)\,\frac{z}{f}, \quad z = D'(u, v). \tag{2}$$

Each surfel is parameterized by a 12-dimensional vector:

$$\mathbf{s} = [\,\mathbf{p},\ \mathbf{n},\ s_u, s_v,\ \alpha,\ \mathbf{c}_{\text{sh}}\,] \in \mathbb{R}^{12}, \tag{3}$$

where $\mathbf{p} \in \mathbb{R}^3$ is the center, $\mathbf{n} \in \mathbb{R}^3$ is the surface normal, (s_u, s_v) are tangent-plane scales, α is opacity, and $\mathbf{c}_{\text{sh}}$ is the 0th-order spherical-harmonic color (higher orders are zero-initialized). Normals are initialized toward the camera direction $(0, 0, 1)$ with small Gaussian perturbation ($\sigma = 0.05$) on the tangential components to emulate brushstroke direction variation. Scales are initialized to the median inter-pixel 3D distance. The surfel cloud is optimized for 3,000 iterations with photometric, normal-consistency, and depth-smoothness losses using the 2DGS framework Huang *et al.* [2024]. Single-view reconstruction is constrained to a $\pm 30°$ explorable viewpoint range.

TripoSplat foreground generation. For each foreground layer (e.g., a human figure), the painting is cropped to the layer's bounding box and TripoSplat Wang *et al.* [2024] generates a set of 3D Gaussians, each parameterized by position, diagonal scale

(s_x, s_y, s_z), and color. The foreground object is rescaled to match its depth-inferred metric size via

$$\lambda_i = \frac{d_i \cdot h_i^{\text{px}}}{f}, \tag{4}$$

where d_i is the object's depth, h_i^{px} its pixel height, and f the focal length.

3DGS→2DGS conversion. To unify the representation, we convert each 3D Gaussian to a 2DGS surfel via covariance eigenvalue decomposition. With a diagonal-covariance simplification, the *shortest scale axis* is taken as the surface normal direction, and the remaining two axes become the tangent-plane scales:

$$k^* = \arg\min_{k \in \{x,y,z\}} |s_k|, \quad \mathbf{n} = \mathbf{e}_{k^*}, \quad (s_u, s_v) = (s_j, s_l)_{j,l \neq k^*}. \tag{5}$$

Colors are unified to linear RGB with SH-DC initialization.

Scene assembly. The background/midground surfel PLY and all foreground surfel PLYs are merged by depth-ordered concatenation into a unified scene $\mathcal{S} = \{\mathbf{s}_j\}_{j=1}^{N}$, which is rendered via 2DGS differentiable rasterization in subsequent stages.

3.3 Stage 3: Skeleton Binding and Constrained Optimization

Skeleton fitting. From the skeleton line-art c_{skel} and the depth map, 2D joint positions are back-projected to 3D using the layer's median depth. For human subjects, we fit a SMPL-X body model Pavlakos *et al.* [2019] by minimizing the Chamfer distance between the SMPL-X mesh and the 2DGS surfel positions. For non-human subjects, a generic articulated skeleton with user-specified topology is used. Single-view depth ambiguity is resolved by user-annotated joint depths on the skeleton drawing.

Gaussian–bone soft-binding network. We introduce a learnable MLP $\mathcal{B}_\theta$ that assigns each surfel $\mathbf{s}_j$ a soft weight vector $\mathbf{w}_j \in \mathbb{R}^K$ over K bones. The input is a 9-dimensional feature vector concatenating the surfel center, the nearest bone-segment midpoint, and the bone direction:

$$\mathbf{x}_j = [\,\mathbf{p}_j;\ \mathbf{m}_{k^*};\ \mathbf{d}_{k^*}\,] \in \mathbb{R}^9, \quad k^* = \arg\min_k \|\mathbf{p}_j - \mathbf{m}_k\|. \tag{6}$$

The network architecture is $9 \to 128 \to 64 \to K$ with ReLU activations on hidden layers and a softmax output:

$$\mathbf{w}_j = \text{softmax}\big(\mathcal{B}_\theta(\mathbf{x}_j)\big), \quad \sum_{k=1}^{K} w_{jk} = 1. \tag{7}$$

Weights are He-initialized and *warm-started* with inverse-distance weights $w_{jk} \propto 1/\|\mathbf{p}_j - \mathbf{m}_k\|$ via 50 steps of cross-entropy gradient descent. A key design principle is that the binding network *constrains motion degrees of freedom rather than driving deformation*: color, opacity, and in-plane scale remain inherited from the 2DGS scene, so the painting's appearance is preserved during motion.

Four-term constrained optimization. After binding, the 2DGS scene is re-optimized with a weighted sum of four losses:

$$\mathcal{L} = \mathcal{L}_{\text{photo}} + \lambda_1 \mathcal{L}_{\text{normal}} + \lambda_2 \mathcal{L}_{\text{skel}} + \lambda_3 \mathcal{L}_{\text{style}}, \tag{8}$$

where:

- $\mathcal{L}_{\text{photo}}$ is the photometric loss between the rendered image and the original painting;
- $\mathcal{L}_{\text{normal}} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \|\mathbf{n}_i - \mathbf{n}_j\|$ enforces normal consistency between adjacent surfels (edge set $\mathcal{E}$);
- $\mathcal{L}_{\text{skel}}$ is the Chamfer distance between bound surfel positions and their associated bone positions, enforcing skeleton-motion adherence;
- $\mathcal{L}_{\text{style}}$ is the negative CLIP Radford *et al.* [2021] cosine similarity between the rendered frame and the original painting, preserving painterly style in CLIP feature space.

The loss weights are $\lambda_1 = 0.1$, $\lambda_2 = 0.5$, $\lambda_3 = 0.05$, reflecting the design priority: skeleton adherence is the strongest constraint (the binding network's primary purpose), normal consistency is moderate, and style is a gentle regularizer. The optimizer is AdamW with cosine learning-rate decay from 10^{-3} to 10^{-4} over 500 epochs.

3.4 Stage 4: Camera Trajectory and Coarse Rendering

Camera trajectory estimation. For each keyframe line-art $c_{\text{kf}}^{(t)}$, the 2DGS scene is rendered from candidate camera poses and the rendered silhouette is matched against the keyframe contour. The R^3 model Xu *et al.* [2025] estimates the relative pose between front and back frame renderings; long trajectories are handled by the `r3_long` variant. We sample 60 candidate poses per keyframe and smooth the resulting trajectory with B-spline interpolation to ensure temporally continuous camera motion.

Coarse rendering. The 2DGS scene is rendered along the trajectory via differentiable rasterization, producing a coarse RGB sequence $\{I_{\text{coarse}}^{(t)}\}$ and a depth sequence $\{D_{\text{coarse}}^{(t)}\}$. To suppress diffusion-model ghosting artifacts in the subsequent stage, depth maps are post-processed with a guided filter He *et al.* [2024] using the RGB frame as the guide:

$$D_{\text{gf}}^{(t)} = \text{GuidedFilter}\big(I_{\text{coarse}}^{(t)},\ D_{\text{coarse}}^{(t)},\ r,\ \varepsilon\big), \quad r = 4,\ \varepsilon = 0.01. \tag{9}$$

The frame rate is 24 fps and the frame count is aligned to $8N + 1$ (LTX-Video's temporal constraint).

3.5 Stage 5: LTX-Video IC-LoRA Style Generation

IC-LoRA Union Control. We use LTX-Video Lightricks Team [2025] with a single IC-LoRA Union Control checkpoint that supports depth, Canny, and pose conditioning through the ComfyUI `LTXAddVideoICLoRAGuide` node. The base model is `ltxv-2b-v0.9.5` with the Euler sampler, 30 steps, and cfg = 1.0.

Two-stage generation. Generation proceeds in two stages:

- **Stage 5a (low-resolution IC-LoRA generation):** resolution 768 × 512, IC-LoRA enabled (`strength_model`= 1.0), conditioned on both the guided-filtered depth sequence (guide strength 1.0) and the coarse RGB sequence (guide strength 0.6). Depth is the primary structural condition; coarse RGB is the secondary appearance condition.
- **Stage 5b (upsampling):** target resolution, IC-LoRA disabled (`strength_model`= 0.0), depth guide disabled, coarse RGB guide retained at strength 0.6 with image-to-image denoising strength 0.4. This refines the low-resolution output to the target resolution while preserving the Stage 5a style.

First-frame latent anchoring. To stabilize temporal style coherence without modifying the diffusion model, we adopt a lightweight anchoring scheme: the first frame's latent code serves as an anchor, and subsequent frames' initial noise is set to the first-frame latent plus a temporal perturbation. This ensures that the style of the first frame—which is closest to the original painting—propagates through the sequence.

ComfyUI workflow. The full workflow chains video loaders, VAE encoding, two sequential IC-LoRA guide nodes (depth at strength 1.0, then coarse RGB at strength 0.6), the LTX video sampler with IC-LoRA enabled (Stage 5a), and a second sampler pass with IC-LoRA disabled and denoising strength 0.4 (Stage 5b), finally combining frames into the output video.

3.6 Complementary Screenplay-Driven Pipeline

While OilPaintingRevive animates a *single* existing painting, the complementary OilPaintingVideo pipeline generates *multi-shot narrative* oil-painting short films from text themes. It comprises four modules:

LLM screenplay generation. A large language model is prompted as a "short-film director screenwriter" to produce a structured JSON screenplay from a user topic, style preset, target duration, and aspect ratio. The system prompt enforces oil-painting aesthetic constraints (composition, lighting, brushstroke, tone), requires each shot to be 2–5 seconds, and specifies camera motion from a fixed vocabulary (`static`, `pan_left/right`, `zoom_in/out`, `tilt_up/down`). Crucially, style words are *excluded* from shot prompts and re-injected deterministically from the style preset during keyframe generation, making style swappable without regenerating the screenplay. The LLM also selects the I2V model per shot (Wan2.1 for high-motion shots, LTX-Video for efficiency). JSON parse failures are fed back to the LLM as self-correction (up to 3 rounds).

Keyframe generation. Each shot's prompt, combined with the style prefix and quality tags, is fed to an SDXL text-to-image pipeline (DPM++ 2M sampler, Karras schedule, cfg = 7.0–7.5, 30–32 steps) with an oil-painting LoRA (strength 0.8) to produce a keyframe image.

Algorithm 1 OilPaintingRevive Pipeline

Require: Painting P, control line-art $C = \{c_{\text{edge}}, c_{\text{skel}}, c_{\text{kf}}\}$
Ensure: Stylized video V

Stage 1: $\{M_i\} \leftarrow \text{SAM2}(P)$ {semantic masks}
$D' \leftarrow \text{LayerNorm}(\text{DA-V2}(P), \{M_i\})$ {Eq. 1}
Parse C into edge, skeleton, keyframe signals
Stage 2: $\mathcal{S}_{\text{bg}} \leftarrow \text{2DGS-Init}(P, D', \{M_i\}_{\text{bg,mid}})$
$\{\mathcal{S}_{\text{fg}}\} \leftarrow \text{TripoSplat}(P, \{M_i\}_{\text{fg}}) \xrightarrow{\text{Eq. 5}} \text{2DGS}$
$\mathcal{S} \leftarrow \text{Assemble}(\mathcal{S}_{\text{bg}}, \{\mathcal{S}_{\text{fg}}\})$
Stage 3: $\mathcal{K} \leftarrow \text{SkeletonFit}(c_{\text{skel}}, D')$ {SMPL-X or generic}
$\mathbf{W} \leftarrow \text{BindingNet}(\mathcal{S}, \mathcal{K})$ {warm-started MLP}
Optimize $\mathcal{S}$ with $\mathcal{L}$ (Eq. 8) {4-loss, 500 epochs}
Stage 4: $\{\mathbf{T}_t\} \leftarrow \text{R}^3\text{Trajectory}(c_{\text{kf}}, \mathcal{S})$
$\{I^{(t)}_{\text{coarse}}, D^{(t)}_{\text{gf}}\} \leftarrow \text{2DGS-Render}(\mathcal{S}, \{\mathbf{T}_t\})$
Stage 5: $V_{5\text{a}} \leftarrow \text{LTX-ICLoRA}(I_{\text{coarse}}, D_{\text{gf}})$ {768×512, IC-LoRA on}
$V \leftarrow \text{LTX-I2V}(V_{5\text{a}})$ {full res, IC-LoRA off, denoise 0.4}
return V

Image-to-video animation. Each keyframe is animated via either LTX-Video (cfg = 1.0, 30 steps, long-edge 768, 64-aligned) or Wan2.2-I2V (cfg = 6.0, guidance 5.0, 30 steps, long-edge 1280, 16-aligned), depending on the per-shot model selection. Frame counts are forced to odd values to satisfy the $8N + 1$ constraint.

Automated editing. Clips are normalized (scale, pad, 24 fps, `yuv420p`, `libx264` CRF 20) and concatenated. A mood-matched background music track is mixed at 0.35 volume with ducking, and narration subtitles are burned in via `drawtext`. The result is a complete narrative short film.

Mesh-to-video module. For geometry-controllable animation, a parallel mesh2video pipeline renders a 3D mesh as white "clay" sequences (with depth and normal passes via software rasterization), then stylizes them via Wan2.1 with depth ControlNet (strength 0.7) or LTX-Video with guide conditioning (strength 0.6). A single style image anchors global appearance, while the depth/clay sequence locks cross-frame geometry—achieving explicit style×geometry decoupling.

3.7 Pipeline Algorithm

Algorithm 1 summarizes the full OilPaintingRevive pipeline. Each stage persists its output to disk (checkpoint mechanism), enabling resumption from the last completed stage upon failure. Serial model loading with memory release between stages keeps peak VRAM within the 24 GB budget (Table 4).

4 Experiments

4.1 Experimental Setup

Dataset. We curate a set of famous oil paintings spanning diverse styles and compositions: portraits (Rembrandt self-portraits, Mona Lisa), landscapes (Starry Night, Wheatfield with Crows), and figurative scenes (The Night Watch). Each painting is paired with hand-drawn control line-art (skeleton for foreground figures, keyframes for camera trajectory, edges for geometric guidance). All paintings are publicly available high-resolution scans.

Implementation details. The pipeline is implemented in Python with PyTorch. Stage 1 uses SAM 2.1 (`hiera_large`) and Depth Anything V2 (ViT-L). Stage 2 uses the 2DGS framework Huang *et al.* [2024] for 3,000 iterations and TripoSplat Wang *et al.* [2024] for foreground objects. Stage 3 uses a numpy/PyTorch MLP binding network ($9 \to 128 \to 64 \to K$) optimized with AdamW (cosine LR $10^{-3} \to 10^{-4}$, 500 epochs). Stage 4 uses the R^3 pose estimation model with 60 pose samples per keyframe and guided filtering ($r = 4$, $\varepsilon = 0.01$). Stage 5 uses LTX-Video (`ltxv-2b-v0.9.5`) with IC-LoRA Union Control via the ComfyUI HTTP API. The complementary OilPaintingVideo pipeline uses an LLM for screenplay generation, SDXL with an oil-painting LoRA for keyframes, and LTX-Video/Wan2.2 for I2V animation.

Hardware. Experiments are conducted on a single NVIDIA GPU with 24 GB VRAM. The serial stage scheduling keeps peak VRAM within budget (Table 4). ComfyUI runs with concurrency = 1 to respect the memory ceiling.

Baselines. We compare against: (i) **LTX-Video I2V**—direct image-to-video generation from the painting scan with no 3D reconstruction; (ii) **Wan2.2-I2V**—a stronger open-source I2V baseline; (iii) **2DGS + bilinear upsampling**—3D reconstruction and rendering without diffusion refinement; and (iv) **LivePortrait** Guo *et al.* [2024]—portrait-only animation for the portrait subset.

Metrics. We evaluate three quality dimensions, following the design specification:

- **Brushstroke style consistency (S_{CLIP})**: CLIP Radford *et al.* [2021] cosine similarity between output frames and the original painting (threshold: ≥ 0.75).
- **Skeleton-motion adherence (E_{skel})**: L2 error between output joint trajectories and target trajectories, in pixels, normalized by the image diagonal (threshold: $\leq 3\%$).
- **Depth-structure fidelity (E_{depth})**: relative L1 error between output depth and coarse 2DGS depth (threshold: ≤ 0.15).

We additionally report LPIPS Zhang *et al.* [2018] perceptual distance and a user study preference rate.

Method	$S_{\text{CLIP}} \uparrow$	$E_{\text{skel}} \downarrow$	$E_{\text{depth}} \downarrow$	LPIPS↓	3D?
LTX-Video I2V	0.78	8.2%	0.31	0.42	✗
Wan2.2-I2V	0.81	6.5%	0.24	0.36	✗
2DGS + Bilinear	0.62	1.8%	0.08	0.55	✓
LivePortrait†	0.74	4.1%	—	0.38	✗
OilPaintingRevive	**0.83**	**2.1%**	**0.12**	**0.31**	✓

Table 1: Main quantitative results. †LivePortrait applies only to the portrait subset. "3D?" indicates whether the method supports novel-view exploration. Our method is the only one that achieves both 3D explorability and high style fidelity.

4.2 Quantitative Results

Table 1 reports the main comparison. OilPaintingRevive achieves the best balance across all three dimensions.

Style consistency. OilPaintingRevive achieves $S_{\text{CLIP}} = 0.83$, surpassing both I2V baselines. This is because the IC-LoRA stage is conditioned on the *original painting's* coarse RGB and depth, providing a strong style anchor. Pure 2DGS rendering ($S_{\text{CLIP}} = 0.62$) suffers from style degradation under novel views, confirming the necessity of the diffusion refinement stage.

Skeleton adherence. Our method achieves $E_{\text{skel}} = 2.1\%$, well below the 3% threshold, and substantially better than I2V baselines (6.5–8.2%) which have no explicit motion control. The 2DGS baseline achieves the lowest error (1.8%) because it directly follows the skeleton, but at the cost of poor style fidelity.

Depth fidelity. OilPaintingRevive achieves $E_{\text{depth}} = 0.12$, within the 0.15 threshold. I2V baselines exhibit high depth error (0.24–0.31) because they generate depth implicitly and often violate the painting's geometry. The 2DGS baseline has the lowest depth error (0.08) since depth is rendered directly from the 3D scene.

LPIPS. OilPaintingRevive achieves the lowest LPIPS (0.31), indicating the highest perceptual similarity to the original painting, confirming that the two-stage IC-LoRA generation effectively preserves painterly appearance.

4.3 Ablation Studies

Loss components. Table 2 ablates the four loss terms in Stage 3. Removing any term degrades performance: $\mathcal{L}_{\text{skel}}$ has the largest impact on skeleton adherence (E_{skel} rises from 2.1% to 7.3%), while $\mathcal{L}_{\text{style}}$ primarily affects style consistency (S_{CLIP} drops from 0.83 to 0.76). The full four-term objective achieves the best overall balance.

Two-stage generation. Table 3 compares the two-stage IC-LoRA strategy against single-stage alternatives. Stage 5a alone (low-resolution) has good style fidelity but poor resolution. Stage 5b alone (full resolution, IC-LoRA on) achieves good style but requires significantly more compute. The two-stage approach (5a→5b) achieves the best quality-efficiency trade-off. Disabling

Configuration	$S_{\text{CLIP}} \uparrow$	$E_{\text{skel}} \downarrow$	$E_{\text{depth}} \downarrow$
Full (4 losses)	**0.83**	**2.1%**	**0.12**
w/o $\mathcal{L}_{\text{photo}}$	0.80	2.4%	0.14
w/o $\mathcal{L}_{\text{normal}}$	0.82	2.3%	0.13
w/o $\mathcal{L}_{\text{skel}}$	0.81	7.3%	0.14
w/o $\mathcal{L}_{\text{style}}$	0.76	2.2%	0.12

Table 2: Ablation of Stage 3 loss components.

Configuration	$S_{\text{CLIP}} \uparrow$	LPIPS↓	Time (s)↓
Stage 5a only (768×512)	0.82	0.35	45
Stage 5b only (full res, IC-LoRA on)	0.83	0.33	180
Two-stage (5a→5b)	**0.83**	**0.31**	90
w/o latent anchoring	0.82	0.39	90

Table 3: Ablation of Stage 5 generation strategy.

first-frame latent anchoring increases temporal flicker (LPIPS rises from 0.31 to 0.39).

Layer decomposition. Removing the SAM 2 layer decomposition (Stage 1) and treating the painting as a single flat layer degrades depth fidelity (E_{depth} rises from 0.12 to 0.21) and increases foreground–background bleeding artifacts, confirming that semantic layering with depth normalization is essential for single-view 3D reconstruction.

4.4 VRAM Budget and Efficiency

Table 4 reports the peak VRAM per stage. Serial model loading with `torch.cuda.empty_cache()` between stages keeps the peak within 24 GB, making the pipeline deployable on a single consumer GPU.

Stage	Peak VRAM	Models
S1: Input parsing	∼6 GB	SAM 2 + Depth Anything V
S2: 3D construction	∼16 GB	2DGS + TripoSplat
S3: Binding optimization	∼12 GB	2DGS + Binding MLP
S4: Trajectory rendering	∼4 GB	2DGS rasterizer + R^3
S5: Style generation	∼24 GB	LTX-Video (19B) + IC-LoR

Table 4: Peak VRAM budget per stage. Serial scheduling ensures the peak never exceeds 24 GB.

4.5 User Study

We conducted a user study with 20 participants (art students and museum visitors). Each participant viewed pairs of videos generated by OilPaintingRevive and the best I2V baseline (Wan2.2-I2V) and selected the preferred one on three criteria: *style fidelity* (how closely it resembles the original painting), *motion naturalness* (how plausible the motion appears), and *explorability* (sense of 3D space). OilPaintingRevive was preferred 78% for style fidelity, 65% for motion naturalness, and 92% for explorability, confirming the advantage of the 3D-explorable hybrid approach.

4.6 Screenplay-Driven Short Films

The complementary OilPaintingVideo pipeline was tested with 15 text themes across three style presets (Van Gogh, Rembrandt, Monet). Each theme produced a 15–30 second multi-shot short film with 5–10 shots. The LLM screenplay generation achieved a 93% JSON parse success rate (first attempt), rising to 100% with one self-correction round. Per-shot generation time averaged 25 s (keyframe) + 90 s (I2V), with full film assembly (including BGM and subtitles) completing in under 20 minutes. The mesh2video module successfully animated 3D mesh assets (e.g., sculptural busts) into oil-painting-style orbit videos using both Wan and LTX backends.

5 Conclusion

We presented *Canvas in Motion*, a hybrid framework that animates famous oil paintings through a principled three-role decomposition: 2D Gaussian Splatting extracts layered 3D structure, hand-drawn skeleton art controls motion via a learnable soft-binding network, and two-stage LTX-Video IC-LoRA diffusion anchors painterly style. The resulting five-stage OilPaintingRevive pipeline achieves CLIP style consistency of 0.83, skeleton adherence of 2.1%, and depth fidelity of 12%—the only method in our comparison that simultaneously provides 3D explorability and faithful style preservation. The complementary OilPaintingVideo pipeline extends this capability to screenplay-driven multi-shot narrative short films with automated editing, demonstrating the generality of the paradigm.

Limitations. Single-view 2DGS reconstruction constrains exploration to ±30°; wider coverage requires multi-view inpainting. Skeleton binding handles articulated subjects but not amorphous phenomena (water, smoke, fabric). Evaluations are on a curated set; a larger benchmark covering more painting genres and art traditions would strengthen generality.

Future work. Directions include: (i) multi-canvas scene composition for cross-painting narratives; (ii) audio-visual alignment for lip-synced talking-painting experiences; (iii) real-time interactive streaming via sparse 2DGS rendering; and (iv) incorporating art-theoretic priors—fat-over-lean layering, focal-point choreography, blank-space regularization—as structured conditioning schedules for the diffusion stage.

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